

Executive Analysis & Management Report

Analytical period: May 2019 – May 2024 admissions

Final analytical population: 54,966 records

1. Executive Summary

This analysis evaluates 54,966 cleaned patient records across clinical conditions, demographics, treatment billing, admission patterns, insurance coverage, medication distribution, length of stay and test-result status. The findings indicate a highly balanced patient mix across major categories, relatively stable average treatment costs over time, and a broadly even split between elective, urgent and emergency admissions.

From a management perspective, the most important signals are: Arthritis is the most frequently recorded medical condition; average billing is approximately \$25.6K per patient and remains within a comparatively narrow range across years; emergency admissions represent 32.9% of the patient population; and the highest-cost 10% of patients account for 5,497 cases requiring closer financial review. The emergency-capacity analysis also identifies July as the highest-volume month for emergency admissions in the period aggregated by calendar month.

The analysis further shows that the five insurance providers have a near-balanced share of patients, while medication utilization is also nearly evenly distributed across the five recorded medications. This reduces the likelihood that a single payer or medication category is driving the overall patient mix and supports a portfolio-based approach to operational planning.

Management Snapshot

KPI	Result	Management interpretation
Final patient records	54,966	Core analytical population after duplicate removal
Average billing amount	\$25,594.53	Overall average treatment cost per record
Median billing amount	\$25,542.75	Average and median are close, indicating limited central skew
Average age	51.5 years	Patient population spans ages 13–89
Average length of stay	15.5 days	Typical stay is approximately 15–16 days
Emergency admissions	18,102 (32.9%)	One-third of admissions require emergency capacity
Top medical condition	Arthritis: 9,218 (16.8%)	Largest recorded condition category
High-cost population	5,497 (10.0%)	Top decile identified for financial monitoring

2. Data Quality & Preparation

The source dataset contained 55,500 records and 15 original fields. No null values were identified in the original columns. A total of 534 duplicate rows were removed, reducing the analytical population to 54,966 records—a reduction of approximately 1.0%.

Billing amounts below zero were treated as invalid and replaced with the overall billing mean used in the cleaning step. Patient names were standardized, admission and discharge dates were converted to date format, and length of stay was derived from the difference between discharge and admission dates.

Data-quality measure	Before cleaning	After cleaning
Records	55,500	54,966
Duplicate rows	534	0
Missing values	0	0
Original fields	15	16 (after derived fields)
Date fields	Text	Datetime
Length of stay	Not available	Derived in days

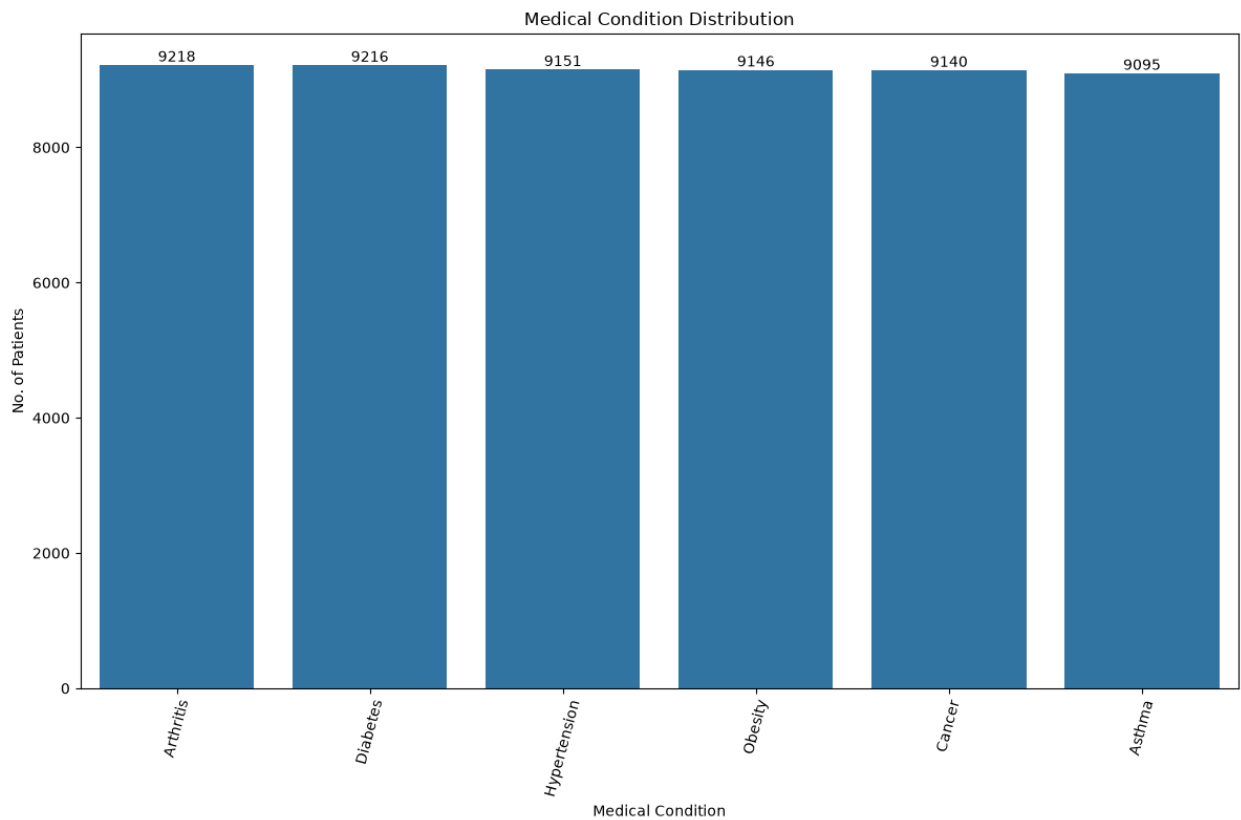
Analytical note

The cleaning process materially improves consistency, but replacing negative billing values with the mean can compress the lower tail of the cost distribution. For production financial reporting, those records should ideally be traced back to source billing systems and classified as refunds, credits, reversals or data-entry errors before final financial reconciliation.

3. Patient Profile & Clinical Mix

Medical condition distribution

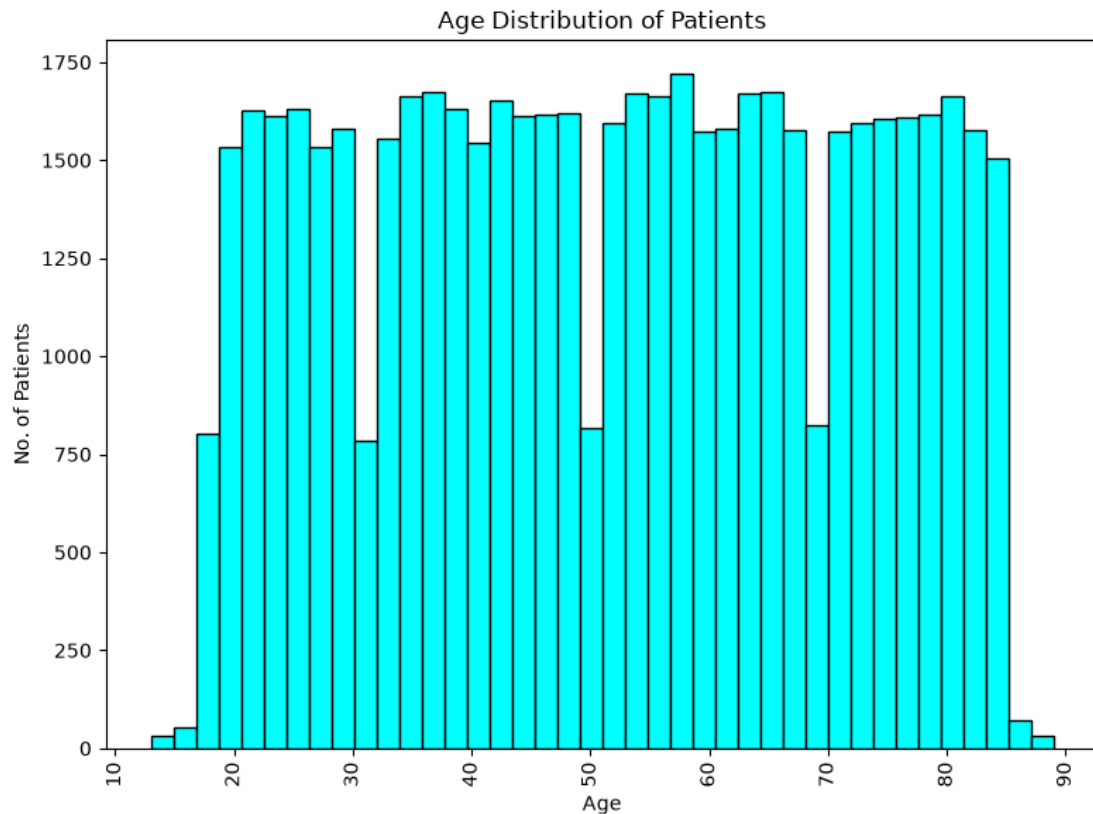
The six recorded medical conditions are closely distributed, with no single condition dominating the population. Arthritis is the largest category at 9,218 patients (16.8%), followed closely by Diabetes at 9,216 (16.8%). The difference between the largest and smallest condition groups is only 123 patients, or about 0.2 percentage points of the full analytical population.



Medical condition	Patients	Share of population
Arthritis	9,218	16.77%
Diabetes	9,216	16.76%
Hypertension	9,151	16.65%
Obesity	9,146	16.64%
Cancer	9,140	16.63%
Asthma	9,095	16.55%

Age profile

Patients range from 13 to 89 years, with a mean age of 51.5 and a median of 52 years. The age histogram shows broad representation across the age spectrum rather than concentration in a narrow age band. This indicates that capacity, clinical pathways and service planning should not be designed around a single dominant age segment.



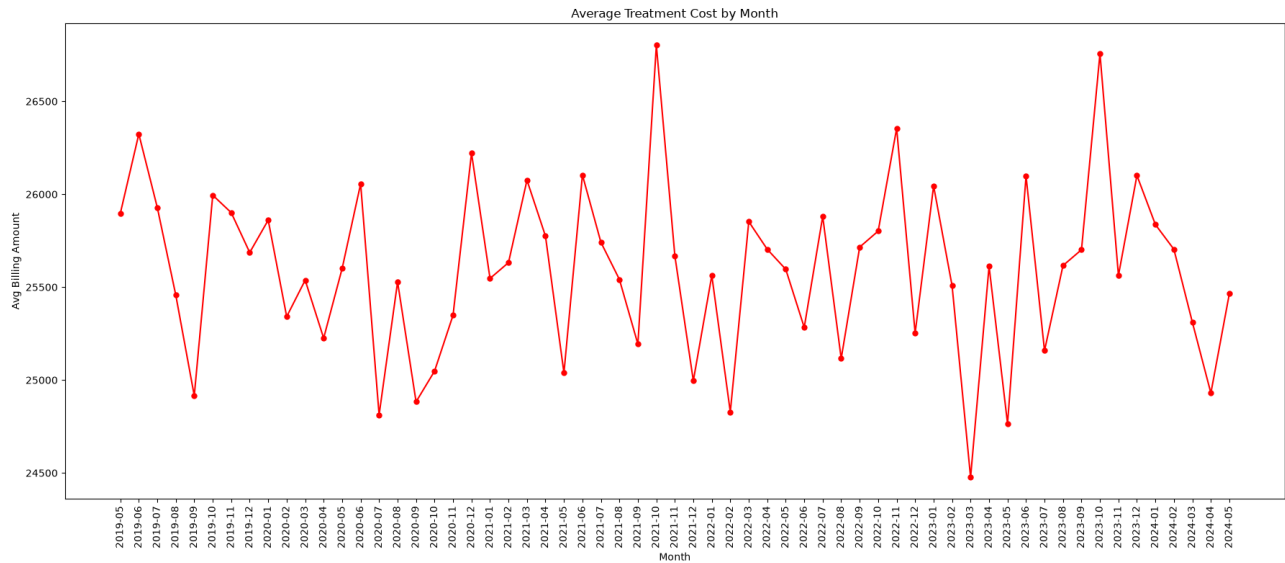
4. Treatment Cost & Utilization Economics

Overall cost position

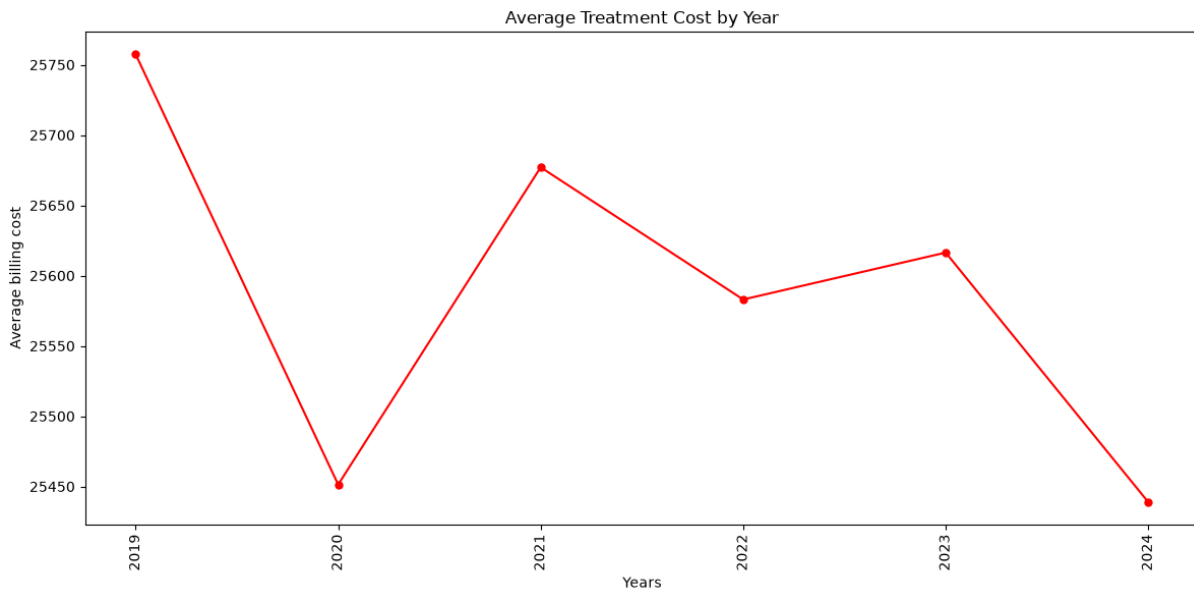
Average billing is \$25,594.53 per patient, with a median of \$25,542.75. The close relationship between these two measures suggests that the central cost level is relatively stable across the population. The interquartile range extends from \$13,326.00 to \$37,819.86, showing meaningful variation in individual bills even though the overall average is stable.

Monthly cost trend

Monthly average billing remains comparatively stable over the 61-month analytical window. The observed monthly values range from approximately \$24.9K to \$26.3K, a spread of roughly 5.6% relative to the lowest month. This supports the interpretation that there is no pronounced month-to-month escalation in average treatment cost.



Annual cost trend



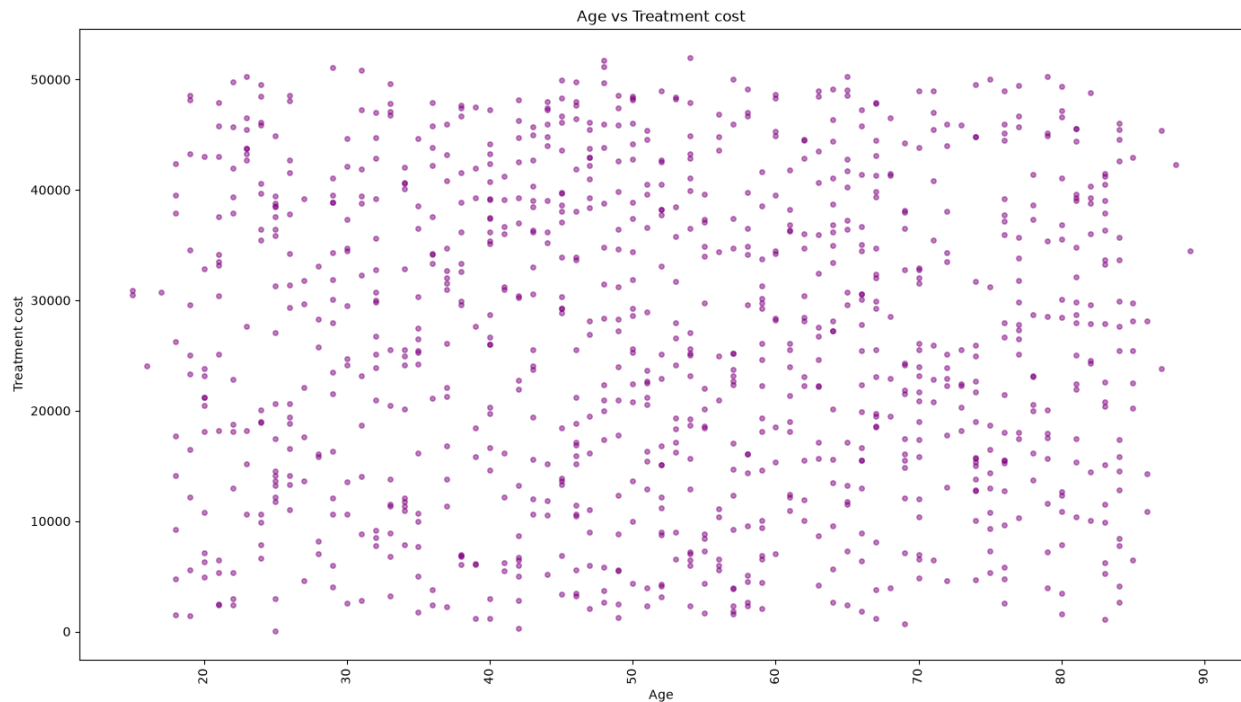
Year	Average billing	Change vs. 2019
2019	\$25,757.42	Baseline
2020	\$25,451.37	-1.19%
2021	\$25,677.04	-0.31%
2022	\$25,583.01	-0.68%
2023	\$25,616.37	-0.55%
2024*	\$25,438.93	-1.24%

*2024 contains admissions through May in the source data; therefore it is not a full-year comparison.

Length of stay and cost

Average length of stay is 15.5 days overall and remains tightly grouped by year: from 15.35 days in 2022 to 15.76 days in 2024. The narrow range indicates that annual changes in average billing are not accompanied by major changes in average stay duration. The analysis therefore does not show evidence of a strong year-level relationship between average length of stay and average billing.

Age versus treatment cost



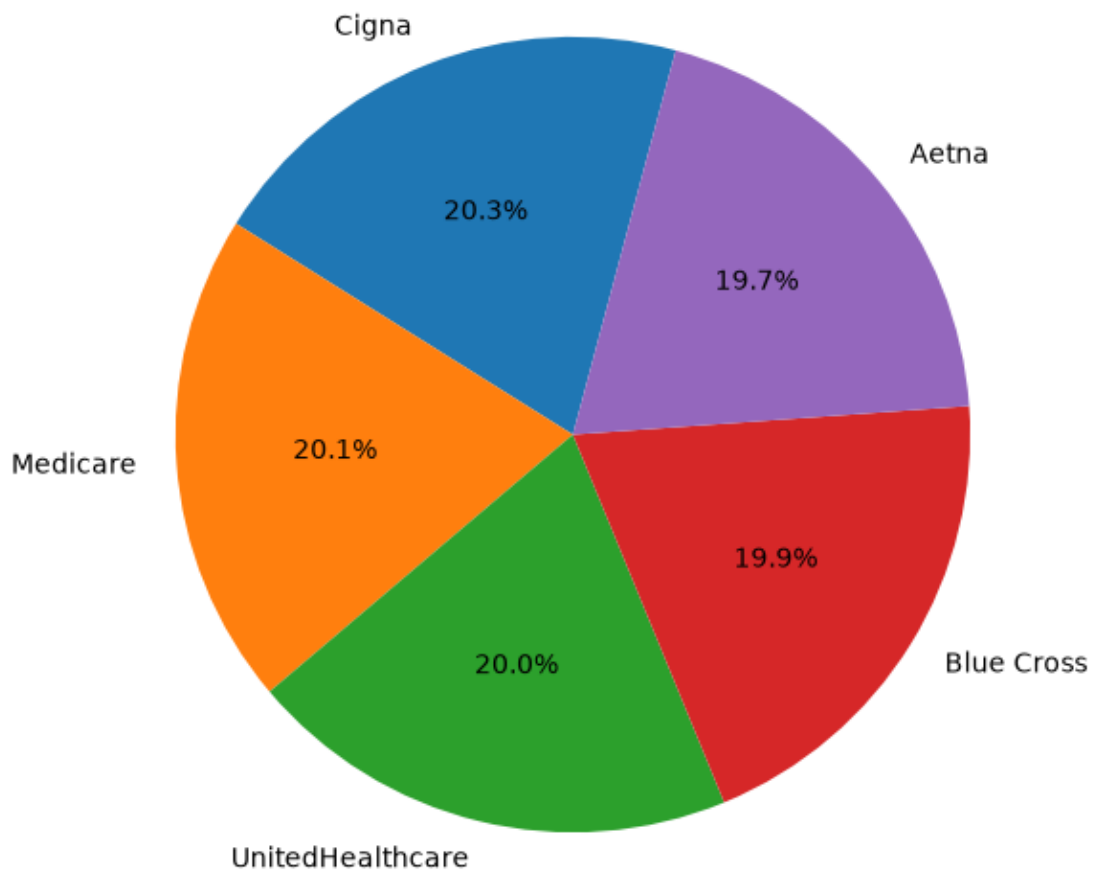
The age-versus-billing scatter plot shows broad dispersion of billing values across ages rather than a clear upward or downward pattern. Within the limits of the sampled visualization, age alone does not appear to explain treatment cost. For a stronger predictive assessment, a correlation coefficient and multivariable model including condition, admission type, medication and length of stay would be appropriate.

5. Insurance Coverage & Admission Mix

Insurance provider distribution

Insurance coverage is highly balanced across the five providers. Cigna represents 20.3% of patients, followed by Medicare at 20.1%, UnitedHealthcare at 20.0%, Blue Cross at 19.9% and Aetna at 19.7%. The largest provider exceeds the smallest by only 0.6 percentage points, indicating that no single payer dominates patient volume.

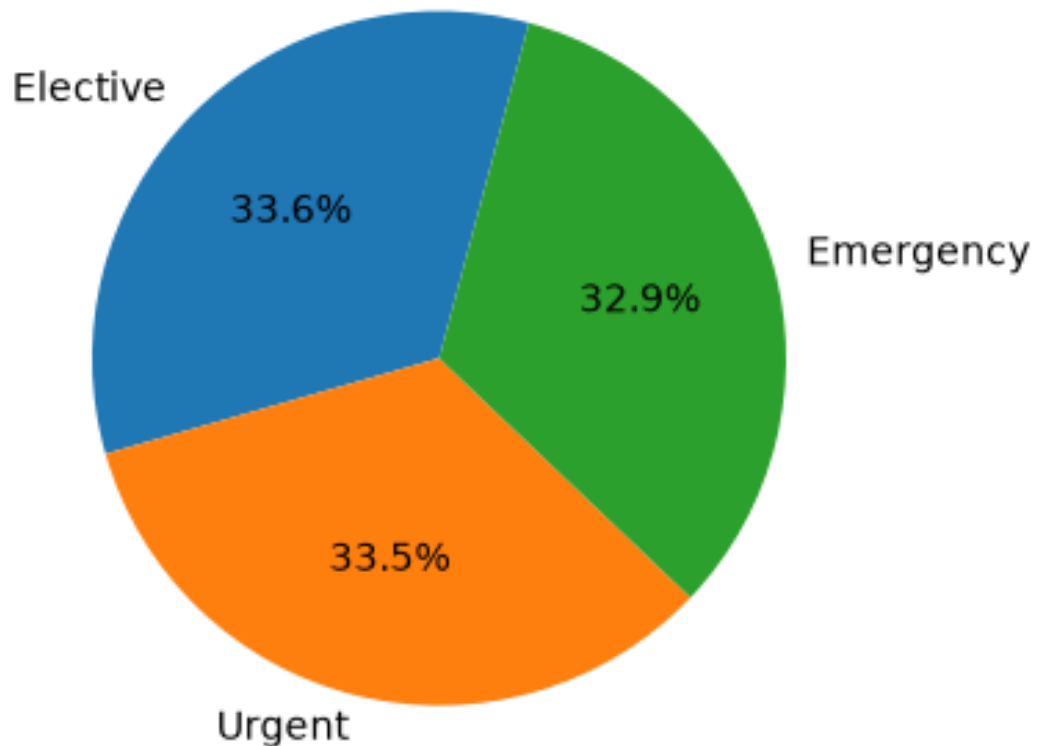
No.of patients in insurance provider



Admission type distribution

Elective admissions account for 18,473 patients (33.6%), urgent admissions for 18,391 (33.5%), and emergency admissions for 18,102 (32.9%). The three categories are therefore almost evenly split, with each contributing roughly one-third of total volume.

Patient Admission type vs patient count



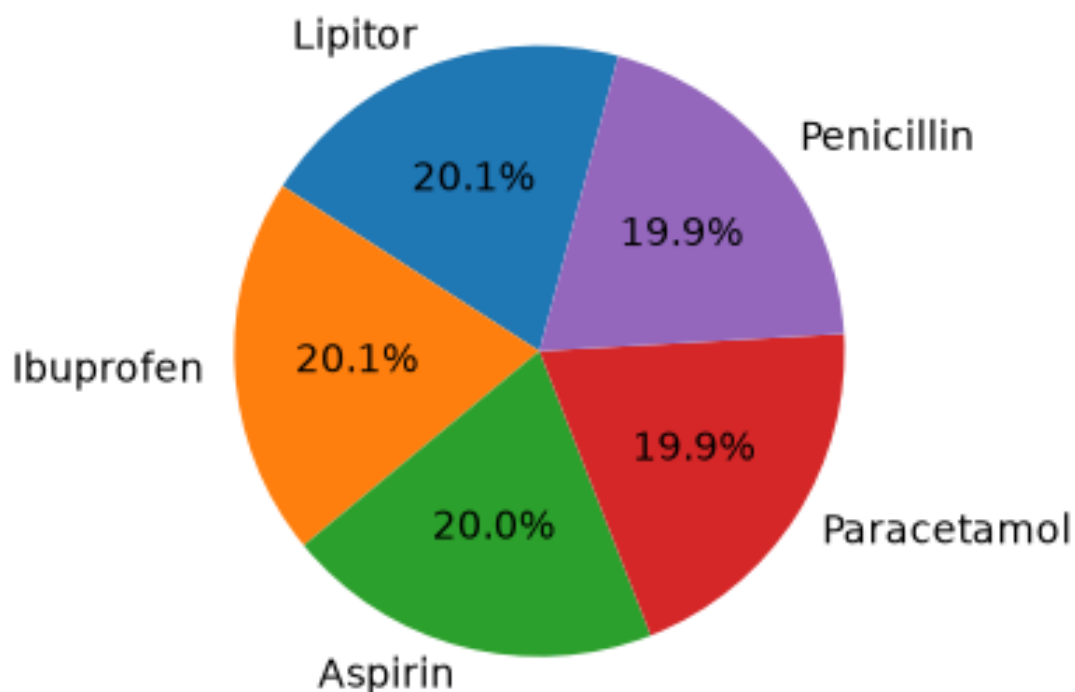
Admission type	Patients	Share
Elective	18,473	33.61%
Urgent	18,391	33.46%
Emergency	18,102	32.93%

6. Medication & Test-result Profile

Medication distribution

Medication utilization is exceptionally balanced. Lipitor is recorded for 11,038 patients (20.08%), followed by Ibuprofen at 11,023 (20.05%), Aspirin at 10,984 (19.98%), Paracetamol at 10,965 (19.95%) and Penicillin at 10,956 (19.93%). The narrow range suggests that medication category alone is not a major volume driver in this dataset.

Patients on Medication



Test-result distribution

Test outcomes are also nearly evenly distributed: 18,437 patients have Abnormal results (33.54%), 18,331 have Normal results (33.35%), and 18,198 are Inconclusive (33.11%). This balance makes test-result status a useful segmentation variable for operational review rather than a rare exception.

7. Operational & Financial Priority Analysis

Emergency department capacity

There are 18,102 emergency admissions, representing 32.9% of the analytical population. When aggregated by calendar month, July records the highest emergency-admission volume at 1,573 cases, followed by January at 1,547 and May/August at 1,533 each. November is the lowest month at 1,462 cases. The difference between the highest and lowest months is 111 admissions, or about 7.6% relative to the November level.

Month	Emergency admissions	Share of annual emergency admissions
January	1,547	8.55%
February	1,452	8.02%
March	1,528	8.44%
April	1,473	8.14%
May	1,533	8.47%
June	1,483	8.19%
July	1,573	8.69%
August	1,533	8.47%
September	1,500	8.29%
October	1,490	8.23%
November	1,462	8.08%

December	1,528	8.44%
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High-cost patient segment

The analysis defines high-cost patients as those above the 90th percentile of billing. This produces 5,497 patients, exactly 10.0% of the final analytical population. This segment should be treated as a financial-review population rather than assumed to represent avoidable or inappropriate care. The next analytical step should be to decompose high-cost cases by medical condition, admission type, length of stay, payer and hospital to identify the operational drivers of elevated spend.

Emergency patients with abnormal results

A targeted review population was created by filtering for patients who were both admitted through the Emergency pathway and recorded an Abnormal test result. This is an appropriate operational prioritization rule because it combines urgency of admission with an adverse test-status flag. The available analysis creates the review list, but the notebook output does not provide a displayed aggregate count for this intersection; therefore no unsupported count is reported here.

8. Management Recommendations

- 1. Build a high-cost case review workflow:** Use the 90th-percentile segment as a recurring financial-monitoring cohort. Break costs down by condition, admission type, length of stay, payer and hospital to identify the sources of cost variation.
- 2. Plan emergency staffing using seasonal volume:** Emergency demand is relatively stable but shows a modest calendar-month spread. Use July as a planning reference point while maintaining capacity across all months rather than relying on a single seasonal spike.
- 3. Segment clinical demand by condition:** The near-even condition mix means that no single condition should dominate capacity assumptions. Service-line planning should consider all six conditions and then prioritize based on cost, acuity and resource intensity.
- 4. Treat payer mix as diversified:** Because the five insurers each represent roughly one-fifth of patients, payer concentration risk appears limited at the patient-volume level. Financial teams should still examine reimbursement differences separately from patient counts.
- 5. Strengthen emergency-result escalation:** Formalize a dashboard or queue for Emergency + Abnormal cases so that review teams can prioritize records using a transparent, reproducible rule.
- 6. Investigate billing anomalies at source:** Negative billing values were corrected during cleaning. Before using the analysis for financial decisions, reconcile these records against source billing transactions to distinguish genuine credits/refunds from data-quality errors.
- 7. Move from descriptive to predictive analytics:** The current analysis establishes patterns. A next-stage model could predict high-cost cases using age, condition, admission type, length of stay, test result, medication, payer and hospital, with model validation before operational use.

10. Executive Conclusion

The healthcare network demonstrates a remarkably balanced patient portfolio across medical conditions, insurers, admission types, medications and test-result categories. Average treatment cost is also comparatively stable over

time, with annual averages remaining within approximately 1.2% of the 2019 baseline. The main management opportunity is therefore not a single dominant volume category, but the identification and explanation of variation within the population—particularly the highest-cost 10%, emergency demand, and Emergency patients with abnormal results.

A practical analytics roadmap is to convert these descriptive findings into recurring management dashboards, with separate views for emergency capacity, high-cost utilization, payer economics and clinical-result prioritization. This would move the analysis from one-time reporting toward continuous operational monitoring.